



Florisynergy

User Manual

Florisynergy IPM — geofenced scouting, spraying and agronomy

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1. About this manual

This manual is organised by job rather than by menu. Find your role in the table below and read that section. Scouts need section 3 only; everything else is for the office.

Section	Who it is for	What it covers
3	Scouts	Walking a round and capturing observations on the phone
4	Supervisors and managers	Reading pressure, acting on recommendations, building and approving sprays
5	Administrators	Setting up blocks, beds, people, chemicals and thresholds
6	Everyone	What the numbers mean, a worked example, and a glossary
7	Everyone	Troubleshooting

1.1 Conventions used

- **Bold** marks something you see on screen — a button, a field or a page name.
- Numbered lists are procedures: follow them in order.
- Coloured panels flag something that is easy to get wrong, or that matters for safety.

2. Getting access

There are two ways into the system and they are meant for different people. Scouts use the phone application. Supervisors and managers use the web portal. A scout account cannot sign in to the portal — this is deliberate.

2.1 Roles and permissions

Role	Signs in to	Can do
Scout	Mobile app	Capture observations; review their own records
Supervisor	Portal	Everything a scout sees, plus recommendations, spray programs and all reports
Administrator	Portal	All of the above, plus farm mapping, workforce and reference data

2.2 Signing in

Everyone signs in with a **device identifier** and a **four-digit PIN** rather than an email address and password. The device identifier ties an account to one phone, which is how the farm knows whose round a set of records belongs to. If a scout changes phone, an administrator updates the identifier under **Settings** → **Workforce**.

LEAVERS

Deactivate an account rather than deleting it. A deleted person takes their history with them; a deactivated one cannot sign in but their past records stay attributed and auditable.

3. For scouts: walking a round

Your job in the app is to record what you find, bed by bed. The app is built to work with no signal — capture the whole round, and it sends itself when you get back into coverage.

3.1 Before you start

- Sign in once. The app remembers you.
- You do not need signal. Walk the round first and worry about sending afterwards.
- Record **every bed you walk**, including the clean ones.

WHY CLEAN BEDS MATTER

The farm's pressure score is the total severity divided by the number of beds scouted. If you record only the beds where you found something, three bad beds look like the whole block is in trouble. Recording the clean ones keeps the numbers honest — and it is the record that proves the block was walked.

3.2 Capturing an observation

1. From the home screen, tap **New scouting**.
2. Under **Location**, choose the **Greenhouse**, then the **Bed / Bay**. The app confirms by GPS that you are inside the block boundary. Where GPS struggles under polythene, scan the greenhouse **QR code** instead.
3. Choose the **Scouting Type** — Pest, Disease, Lure or Sticky Trap. The form changes to suit.
4. Complete the details for that type (see 3.3).
5. Set the **severity** on the 0–5 scale (see 3.4).
6. Add **Notes** where a number and a photograph will not convey it.
7. Add a **Photo** for anything unusual, from the camera or the gallery.
8. Add the entry. It joins the queue for this round.
9. Repeat for each bed, then **submit the round**. Add an overall comment if something affected the whole walk — weather, irrigation running, and so on.

3.3 What each scouting type asks for

Type	You record
Pest	Variety · Pest · Stage (egg, larva, nymph, adult) · Where on the plant · Severity · Beneficials seen

Type	You record
Disease	Variety · Disease · Where on the plant · Severity
Lure	Lure ID · Count caught · Severity
Sticky trap	Trap ID · Count on the card · Severity

3.4 The severity scale

Use the same scale every time. Consistency between scouts matters more than precision, because the system compares this week to last week on the same bed.

Score	Meaning
0	Nothing found. The bed was walked and it is clean.
1	Trace. One or two plants, barely there.
2	Light. Present but not spreading.
3	Moderate. Established, worth acting on.
4	Heavy. Raises an alert on its own, wherever it is found.
5	Severe. Crop at risk.

SEVERITY 4 IS A HARD LINE

Any single reading of 4 or above raises an alert immediately, even where the rest of the block is clean. Do not round down to avoid causing a fuss, and do not round up to make a point — the office is comparing your number against the same bed last week.

3.5 Sending your round

- Records queue on the phone until there is signal, then send themselves.
- Sending twice cannot create duplicates — each record carries its own identifier.
- You can carry on working while a queue is sending.
- Tap any record in your list to review what you captured.

4. For supervisors and managers

The portal is organised as a loop: see the pressure, open the recommendation it raised, build the spray that answers it, then check whether it worked.

4.1 Where things are

Page	What it is for
Dashboard	The morning overview — headline numbers and the farm map
Pressure Map	The estate coloured by pressure; select a block for detail
Scouting	Every observation, filterable; select one for its full record
Recommendations	What the thresholds have raised, and what became of it
Spray Programs	Applications, dosing, cost and pre-harvest compliance
Analytics	Twelve reports across scouting and spray
Settings	Farm mapping, workforce and reference data (section 5)

4.2 Reading a block

1. Open **Pressure Map** and set the date range.
2. Select a block. The panel gives a headline sentence naming the worst active issue.
3. Open **Pressure by pest / disease**. Each agent carries its own index, its own threshold and its own verdict; they are never averaged together.
4. The **Records** tab lists findings first, worst severity at the top. Beds walked with nothing found sit behind a toggle.
5. Select any record to open its full page: the round it belongs to, every earlier reading of that agent on that bed, and what was done about it.

4.3 The recommendations board

Recommendations move left to right through four columns. The board is the record of what the farm decided, not merely what the system suggested.

Column	Meaning
Open	Raised by a threshold breach or a severity-4 hotspot. Nobody has acted.
Planned	Someone has decided to act; the spray has not yet gone out.
Actioned	A spray program has been created against it.
Resolved	A later round found the agent below its threshold.

Outcome verdicts

Once a program has been applied, the next scouting round decides the outcome on its own. One of three verdicts appears on the card:

- **Recovered** — the next reading came in below the threshold.
- **Recovering** — falling, but not yet below the line.
- **Not responding** — still at or above the threshold after the intervention. This is the one to escalate; consider a different mode of action.

RECURRENCE REOPENS AUTOMATICALLY

Where an agent breaches its threshold again on a block that was previously resolved, the recommendation reopens and is marked as a recurrence. Problems that keep returning do not quietly disappear from the board.

4.4 Building a spray program

The engine suggests; a person decides. Nothing is written until you commit it.

1. From an open recommendation, start a program — or use **New program** on the Spray Programs page for a routine application.
2. **Location** — greenhouse, bed or bay, partition, variety, and the scouting report date this application answers.
3. **Application** — type (foliar, drench, fogging, dusting, drip), coverage (full or top), re-entry hours, volume of water, start date and time.
4. **Add a product** — select it and enter the rate per 100 litres. The quantity calculates as you type.
5. Add further products for a tank mix. Each carries its own rate.
6. Read the **Compliance** panel before committing (see 4.5).
7. Check the total cost and the safe-to-harvest date, then **Create program**.

How the quantity is calculated

$$\text{quantity} = \text{volume of water} \times \text{rate} \div 100,000$$

Rate is millilitres or grams of product per 100 litres of water, as written on the spray sheet. A 1,000 litre tank at a rate of 50 needs 0.5 litres of product.

CHANGE THE TANK VOLUME AND EVERYTHING RE-PRICES

Quantity and cost for every product in the mix recalculate when you change the water volume. What you see before committing is what gets saved.

4.5 What the compliance panel checks

Check	Level	Meaning
Resistance rotation	Blocks	The same RAC mode-of-action group has gone onto this block within 28 days

Check	Level	Meaning
Tank-mix conflict	Blocks	Two products in this tank share a RAC group — no resistance benefit
Target fit	Warns	The product is not labelled for the pest or disease being targeted
WHO hazard class	Warns	Class II or above — confirm PPE and authorisation
PHI and REI	Informs	Pre-harvest and re-entry intervals carried onto the record

A blocking issue can be overridden, but you must tick the override box and the reason is written onto the program permanently. That is the audit trail; it is meant to be visible rather than convenient.

4.6 The approval sheet

Every program has a printable one-page authorisation, opened from the Spray Programs page or from **Analytics** → **Programs**. It carries:

- The scouting report date, and what the application is answering.
- Block, bed, partition, variety, application method, coverage, tank volume and block area.
- Every product with its active ingredient, target, WHO class, RAC group, rate, quantity, unit price and line cost, with a total.
- Re-entry hours, safe-to-harvest date, and a hazard warning naming the products.
- Signature blocks for the agronomist, the farm manager and the spray operator.

4.7 Reports worth knowing

Report	Answers
Trends	Severity over time, one line per pest and per disease. Is it improving?
Movement	Select a scout to see their walk bed by bed, with time spent on each
Coverage	Which blocks received full cover and which only a top pass
Cost by chemical	Spend, then quantity used, then how often applied
Programs	Expand a row for the tank mix; exports to CSV and to a printable report

5. For administrators: setting the farm up

Everything configuration-related sits under **Settings**. Work through these in order — the later steps depend on the earlier ones.

5.1 Farm mapping

1. Draw each greenhouse boundary on the satellite map, as a rectangle or a freeform shape. This boundary is what geofences the scouts.
2. Generate the beds. Use bulk generate for “Bed 1 ... Bed N”, or add them individually.
3. Note the QR code for each block and print it for the greenhouse door.

REGISTER EVERY BED

The pressure index divides by the number of beds scouted. A block whose beds are only half registered reports pressure roughly twice as high as it should. This is the most common and most damaging setup error.

5.2 Workforce

- Add each person with a role: scout, supervisor or administrator.
- Give each a device identifier and a four-digit PIN.
- Deactivate rather than delete when someone leaves.

5.3 Reference data

- **Varieties** — the crop varieties scouts can choose from.
- **Pests and diseases** — each with a severity ETL and a pressure ETL.
- **Chemicals** — the register: product, price, WHO class, RAC group, rate, pre-harvest and re-entry intervals.
- **ETL override rules** — tighter thresholds for a particular variety or block, for instance where a market demands it. Every change is recorded with a reason.

5.4 Setting thresholds

Two numbers per pest and per disease, doing different jobs:

Threshold	What it does
Severity ETL	A single reading at or above this raises a recommendation on its own. Typically 2 to 4, depending on how aggressive the agent is.
Pressure ETL	The block-wide index threshold. These are naturally small numbers: one severity-3 bed in a twenty-bed block gives an index of 0.15.

6. What the numbers mean

6.1 The pressure index

For one pest or one disease, in one block, over the chosen date range:

$$\text{pressure index} = \text{total severity across beds} \div \text{beds scouted}$$

- Beds walked with nothing found count as zero, and pull the index down.
- Pests and diseases are scored separately and never averaged together.
- **Action is required when the index reaches the ETL, or any single reading reaches severity 4.** Either condition alone is sufficient.

6.2 Worked example

A twenty-bed block, all twenty walked. Powdery Mildew is found on four beds at severities 4, 3, 2 and 1; the remaining sixteen are clean.

Step	Value
Total severity	$4 + 3 + 2 + 1 = 10$
Beds scouted	20
Pressure index	$10 \div 20 = 0.50$
Pressure ETL for Powdery Mildew	0.20
Verdict	Action required — the index is over ETL, and the severity-4 bed is a hotspot in its own right

6.3 Glossary

Term	Meaning
ETL	Economic threshold level — the point at which acting costs less than not acting
PHI	Pre-harvest interval — days after spraying before the block may be cut
REI	Re-entry interval — hours after spraying before anyone may enter
RAC group	Mode-of-action code. Rotating groups is what prevents resistance
WHO class	Acute hazard classification. Class II and above requires full PPE
Hotspot	Any single observation at severity 4 or above
Program	One spray event, which may contain several tank-mixed products
Round	One scout's walk of a block, submitted together as a batch
Dwell cap	45 minutes. A longer gap between two records is counted as a break, not as time spent on a bed, so the movement report is not inflated by lunch

7. Troubleshooting

Symptom	What to do
A scout's records have not appeared	They are queued on the phone awaiting signal. Open the app in coverage and they send themselves. Sending twice cannot duplicate them.
The app will not accept a location	GPS cannot confirm the scout is inside the boundary, which is common under polythene. Scan the greenhouse QR code instead.
A block reports impossibly high pressure	Almost always missing beds. Check under Settings → Farm Mapping that every bed in that block is registered.
A spray is blocked and should not be	Read the reason given. Where it is resistance rotation, choose a different RAC group — the check is doing its job. Override only with a stated reason.
A record looks wrong	Obvious outliers are flagged automatically against the block's own history and appear marked in the reports. Open the record to see why it was flagged.
The wallboard says data may be stale	It has not refreshed for five minutes. Check the display's network connection, then reload the page.
A cost looks wrong on a program	Check the tank volume and the rate per 100 litres. Quantity, and therefore cost, is derived from both.

If a problem is not listed here, note what you were doing, what you expected and what happened, and pass it to your system administrator with the date and time.